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Aging in Sport and Exercise

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AUDIO FOR FIGURE 18.11 describes the mechanisms contributing to the decline in endurance performance with aging.

Physiological Adaptations to Exercise Training

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ACTIVITY 18.3 *Cardiorespiratory Changes with Aging* shows how your aerobic capacity will change over the next 50 years, depending on your lifestyle.

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In Closing

Dara Torres is the first and only swimmer from the United States to compete in five different Olympic Games (1984, 1988, 1992, 2000, and 2008). More impressively, Torres is the first swimmer from any country to compete in the Olympic Games past the age of 40! On August 17, 2008, at the age of 41 years and 125 days, she won the silver medal in the women's 50 m freestyle, finishing in a new American record time of 24.07 s (a heartbreaking 0.01 s behind the winner). Only half an hour later, she won another silver medal as part of the U.S. 4 × 100 m medley relay team, swimming the fastest 100 m freestyle split (52.27 s) in relay history. She followed that up with yet a third silver medal, this time in the 4 × 100 m freestyle relay, swimming the anchor leg for the United States.

Years of training at such a high level, however, took its toll on her aching shoulders and arthritic knees. Following reconstructive knee surgery and rehabilitation, Torres started training once again in 2010 with the goal of competing in the 2012 London Olympics. At the 2012 U.S. Olympic Trials, at age 45, she placed fourth in her specialty event—the 50 m freestyle. Even so, she was only 0.09 s from qualifying for her sixth U.S. Olympic team. Finally, odds-defying Dara Torres had concluded her illustrious Olympic career.

The number of men and women over age 50 years who exercise regularly or participate in competitive sport has increased dramatically over the past 30 years. According to current population forecasts, the number of elderly people will increase worldwide from 6.9% of the population in 2000 to a projected 19.3% by 2050. In parallel with this overall increase in older adults, the number of middle-aged and older athletes is expected to increase as well. Many of these older competitors, often termed masters or senior athletes, engage in competition for fun, general recreation, and fitness, while others train with the same enthusiasm and intensity as Olympians. Opportunities are now available for older athletes to compete in activities ranging from marathon running to powerlifting. The success achieved and the performance records set by many older athletes are phenomenal. However, although these older athletes exhibit strength and endurance capacities that are far greater than those of untrained people their age, even the most

highly trained older athlete experiences a decline in performance after the fourth or fifth decade of life.

In modern societies, voluntary physical activity begins to decline soon after people reach physical maturity. Technology has made many aspects of life less physically demanding. Voluntary participation in strenuous physical activity on a regular basis is an unusual pattern of behavior that is not observed in most aging laboratory animals. Studies have shown that humans and other animals tend to decrease their physical activity as they grow older. As shown in [figure 18.1](#), rats that were allowed to eat ad libitum ran an average of more than 4,000 m (4,374 yd) per week in the early months of life but covered less than 1,000 m (1,094 yd) per week during their final months.

Considering the importance of exercise for maintaining muscular and cardiorespiratory fitness, it is not surprising that inactivity can lead to deterioration of one's capacity for strenuous effort. Because of this, it is difficult to distinguish between the effects of chronological aging alone (known as **primary aging**) and those of the reduced activity and comorbidities that often accompany aging. Researchers who study aging most commonly use cross-sectional designs, but these have important limitations compared to longitudinal study designs. For example, changes in medical care, diet and exercise, and other lifestyle variables may affect age cohorts differently. Selective mortality—that is, the fact that the subject population consists of the survivors of a cohort that has already experienced some degree of mortality—is an issue as well. Finally, when all but apparently healthy older individuals are excluded from exercise studies, it is difficult to apply research findings to the larger aged population, many of whom have underlying disease, take medications, or both. It is important to understand the impact of primary aging on physiological function, but the interpretation and applicability of the results are influenced by the study design, as well as the specific population being tested.

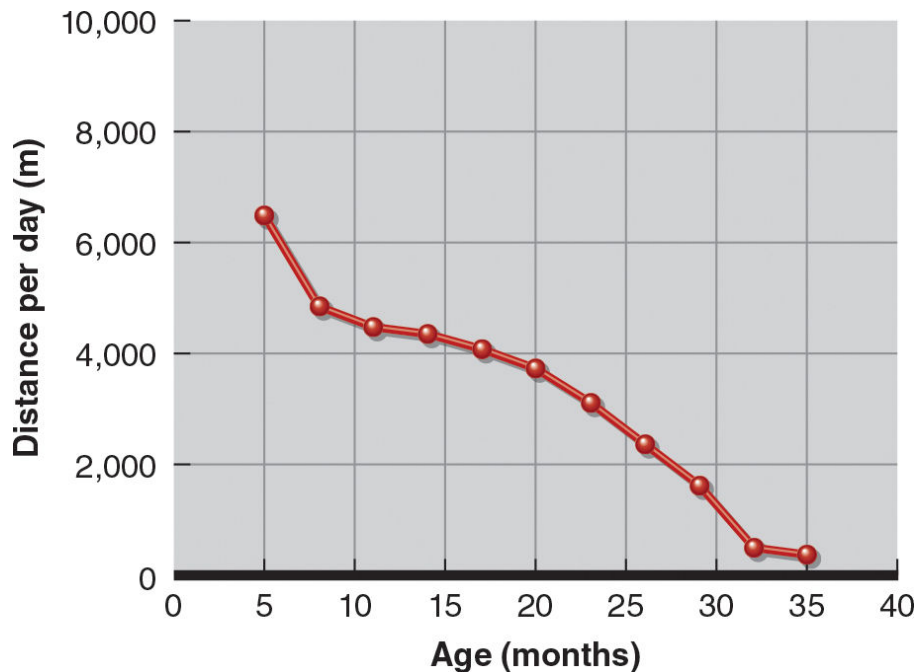


FIGURE 18.1 Voluntary running activity in rats throughout life.

Adapted by permission from J.O. Holloszy, "Mortality Rate and Longevity of Food-Restricted Exercising Male Rats: A Reevaluation," *Journal of Applied Physiology* 82 (1997): 399-403.

Height, Weight, and Body Composition

As we age, we tend to lose height and gain weight, as illustrated in [figure 18.2](#).³⁵ The reduction in height generally starts at about 35 to 40 years of age and is primarily attributable to compression of the intervertebral disks and poor posture early in aging. At about age 40 to 50 years in women, and 50 to 60 years in men, osteopenia and osteoporosis become a factor. **Osteopenia** is a reduction in bone mineral density below normal that occurs before **osteoporosis**, which is a severe loss of bone mass with deterioration of the microarchitecture of bone, leading to increased risk of bone fracture (see [chapter 19](#)). Genetic factors and poor diet and exercise habits throughout the life span contribute to the development of osteoporosis in both men and women, while decreased estrogen concentrations after menopause appear to be responsible for the greater rate of bone loss in women. During the adult life span, a gain in weight typically occurs between age 25 and 45 and is attributable to both a decrease in physical activity levels and excess caloric intake. Beyond the age of 45, weight stabilizes for about 10 to 15

years and then decreases as the body loses bone calcium and muscle mass. Many people over 65 years of age tend to lose their appetite and thus don't consume sufficient calories to maintain body weight. An active lifestyle, however, tends to help stimulate appetite so that caloric intake more closely approximates caloric expenditure, thereby maintaining weight and preventing frailty in old age.

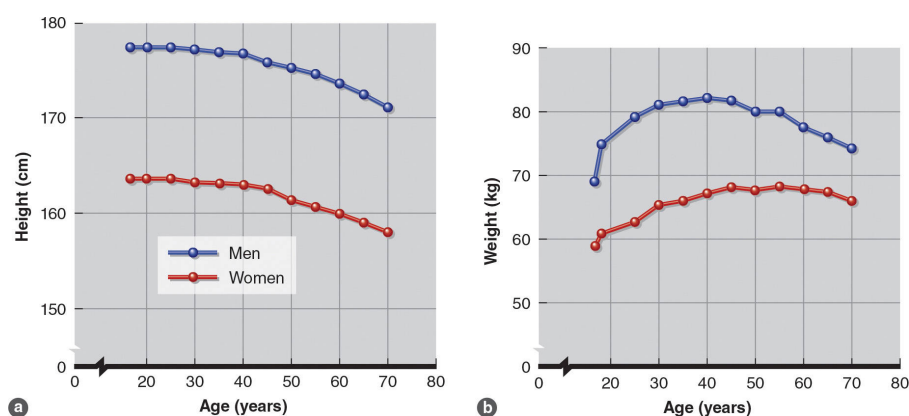


FIGURE 18.2 Changes in (a) height and (b) weight in men and women up to 70 years of age.

Reprinted by permission from W.W. Spirduso, *Physical Dimensions of Aging* (Champaign, IL: Human Kinetics, 1985), 59.

Beginning at about 20 years of age, humans also tend to gain fat. This is largely attributable to changes in diet, physical inactivity, and a reduced ability to mobilize fat stores. However, the body fat content of physically active older people, including older athletes, is significantly lower than that of age-matched sedentary people. In addition, with primary aging there tends to be a shift in the location where body fat is stored, from the periphery toward the center of the body around the organs. This *central adiposity* is associated with cardiovascular and metabolic diseases. Although physical activity cannot fully counteract the age-related gain in fat mass, in active men and women there is less of a shift of the fat stores with aging, which is more advantageous for reducing the risk of cardiovascular and metabolic disease.

Fat-free mass decreases progressively in both men and women beginning at about the age of 40. This results primarily from decreased muscle and bone mass, with muscle having the greatest effect because it constitutes about 50% of the fat-free mass.

Sarcopenia is the term used to describe the loss of muscle mass associated with the aging process. [Figure 18.3](#) illustrates the changes in muscle mass with aging in a cross-sectional study of 468 men and women, aged 18 to 88 years.²¹ There is almost no decline in muscle mass until about age 40, at which time the rate of decline increases, with a greater decline in men than women. Obviously, a reduction in physical activity is a major cause of this decline in muscle mass with aging, but there are other factors. For example, the rate of muscle protein synthesis is reduced while the rate of muscle protein breakdown is unchanged or accelerated with aging, leading to negative nitrogen balance and net loss of muscle. The rate of muscle protein synthesis in 60- to 80-year-olds is about 30% lower than in a 20-year-old. This reduction in muscle protein synthesis rate in older people is likely associated with declines in growth hormone, insulin-like growth factor 1,¹⁴ and cell signaling. Longitudinal data suggest that loss of fat-free mass and gain of fat mass offset each other. As a result, percent fat is increasing while total body mass remains relatively stable.

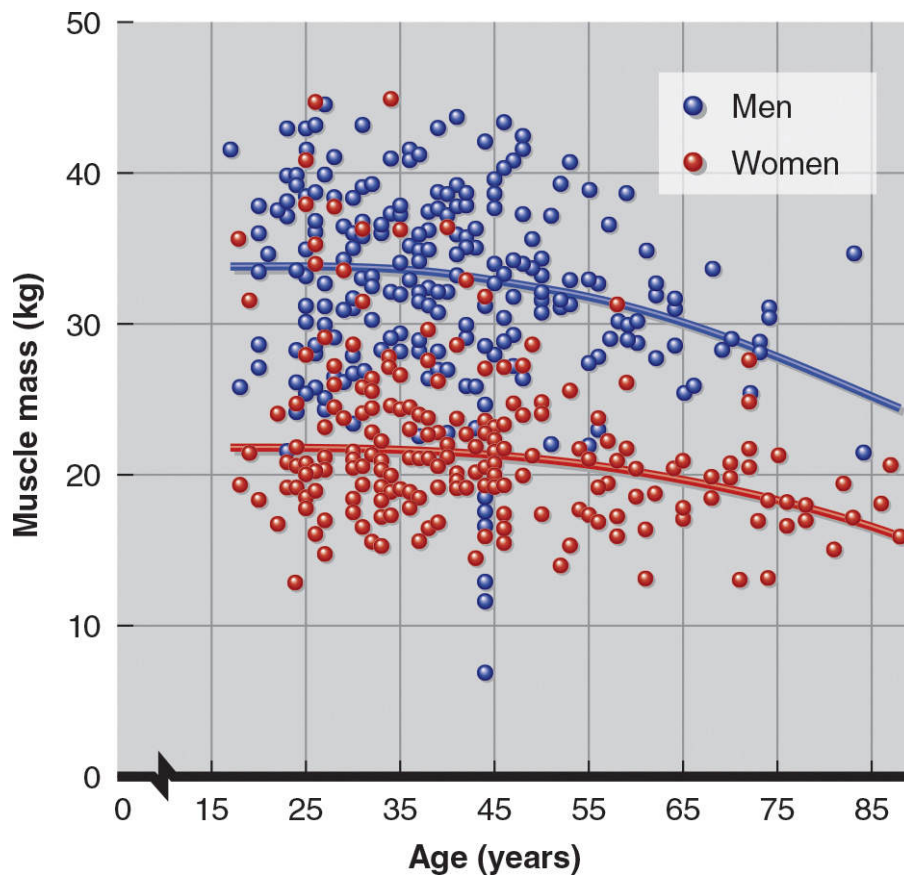


FIGURE 18.3 Changes in muscle mass with aging in 468 men and women 18 to 88 years of age. The rate of decline is greater in men than in women and is steeper after about 45 years of age.

Adapted by permission from I. Janssen et al., "Skeletal Muscle Mass and Distribution in 468 Men and Women Aged 18-88 yr.," *Journal of Applied Physiology* 89 (2000): 81-88.

VIDEO 18.1 Presents Scott Trappe on age-related sarcopenia.



There is also a significant decrease in bone mineral content, starting at about age 30 to 35 in women and at age 45 to 50 in men.

Throughout the life cycle, bone is constantly being formed by osteoblasts and resorbed by osteoclasts. Early in life, resorption occurs at a slower rate than synthesis and bone mass increases. With aging, resorption exceeds synthesis, resulting in a net loss of bone. The loss of both muscle and bone mass is at least partially attributable to decreased physical activity, especially a lack of weight-bearing exercise. Since bone mineral accounts for less than 4% of total body mass in young adults, the contribution of osteopenia to the loss of total fat-free mass is small compared to that of sarcopenia.

These differences in weight, relative (%) body fat, fat mass, and fat-free mass with aging are illustrated in [figure 18.4](#).²⁵ These data are from a study of young (18-31 years) and older (58-72 years) men and women who either were sedentary or were endurance-trained athletes. Body weight, relative body fat, and fat mass were higher in the older sedentary groups, whereas fat-free mass was lower. Similar trends, except for body weight, were noted for the endurance-trained athletes. However, the young and older endurance-trained athletes had much lower total body weight, relative body fat, and fat mass values and similar fat-free mass values compared to their sedentary age-matched counterparts.

With training, older men and women can reduce weight, percent body fat, and fat mass. Furthermore, they can increase their fat-free mass, but as in younger individuals, this is more likely with resistance training than with aerobic training. Men appear to experience greater changes in body composition than women, but the reasons for this have not been clearly established.

For older adults desiring to lose weight and body fat, the most significant changes in body composition result from a combination of diet and exercise, with a modest reduction in caloric intake (500-1,000 kcal/day) being the preferred approach. A more substantial reduction in caloric intake (>1,000 kcal/day) is likely to result in a loss of fat-free mass as well as fat mass. This is not desirable, since a loss in fat-free mass is associated with a reduction in resting metabolic rate, thus decreasing the rate of weight and fat loss. Exercise that increases fat-free mass will likely increase resting metabolic rate, which would increase the rate of weight loss. It

appears that older adults experience changes in body composition due to exercise training that are similar to those for younger adults.

In Review

- Body weight tends to increase with aging, whereas height decreases.
 - Body fat increases with age, primarily because of increased caloric intake, decreased physical activity, and a reduced ability to mobilize fat.
 - Beyond age 45, fat-free mass decreases, primarily because of decreased muscle and bone mass, both resulting at least partly from decreased activity.
 - The sarcopenia observed with aging is thought to be the result of decreased muscle protein synthesis in older people, accompanied by an unchanged or accelerated rate of muscle protein breakdown.
 - Exercise training can help attenuate these changes in body composition, even in individuals as old as 80 to 90 years of age.
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Physiological Responses to Acute Exercise

As we age, muscular and cardiovascular endurance and muscular strength tend to decrease, with the extent of decrease dependent on physical activity and genetics. As activity decreases, which appears to be a natural phenomenon in both animals and humans, these reductions in physiological function are much more substantial.

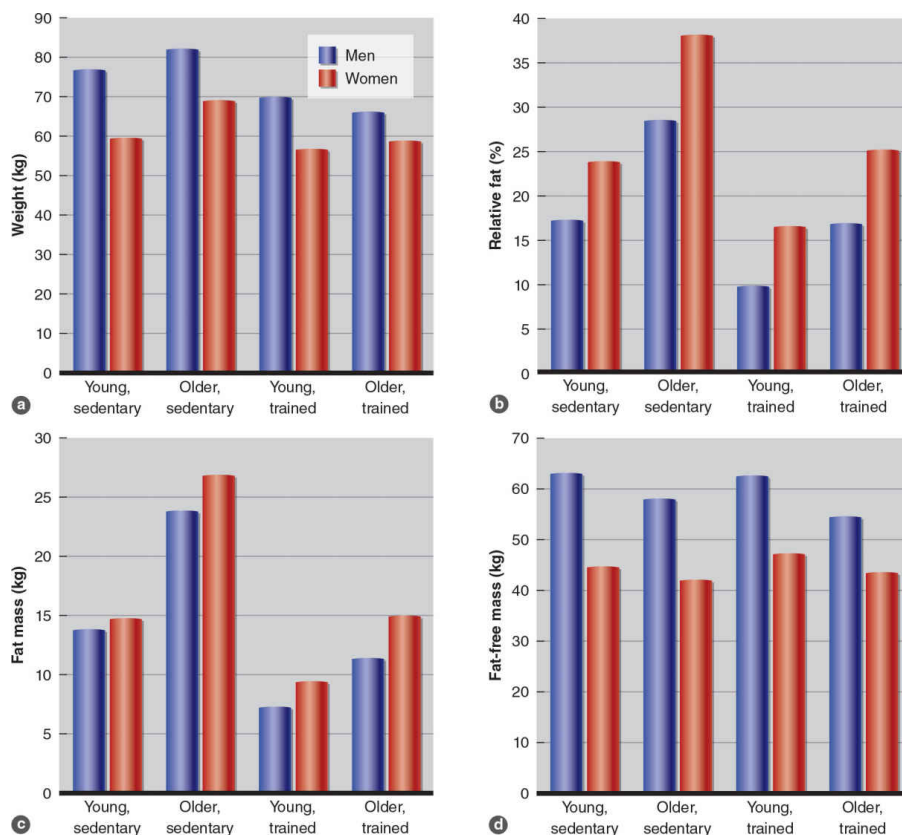


FIGURE 18.4 Differences in (a) total body weight, (b) relative body fat, (c) fat mass, and (d) fat-free mass in young and older men and women, sedentary and endurance trained.

Adapted by permission from W.M. Kohrt et al., "Body Composition of Healthy Sedentary and Trained, Young and Older Men and Women," *Medicine and Science in Sports and Exercise* 24 (1992): 832-837.

Strength and Neuromuscular Function

The strength needed to meet the activities of daily living remains unchanged throughout life. However, a person's maximal strength, generally well above the daily demands early in adulthood, decreases steadily with aging. Eventually, strength may decline to the point where previously simple activities become challenging. For example, the ability to stand up from a sitting position in a chair starts to be compromised at age 50, and before age 80 this task becomes impossible for some people (see [figure 18.5a](#)). As a further example, opening the lid on a jar is a task that can easily be accomplished by almost all men and women below the age of 60. After age 60, the failure rate for this task increases significantly.

[Figure 18.5b](#) shows leg strength changes with aging in men. Knee extension strength in normally active men and women starts to

decrease by age 40. But training the knee extensor muscles with resistance exercises enables older men to perform better at age 60 than most normally active men half that age. The reduction in strength with aging was highly correlated with the reduction in the cross-sectional area of the involved muscles. The reductions in strength with aging appear to be modality specific, in that losses in isokinetic strength are greatest at high angular velocities and losses in concentric strength are greater than losses in eccentric strength.

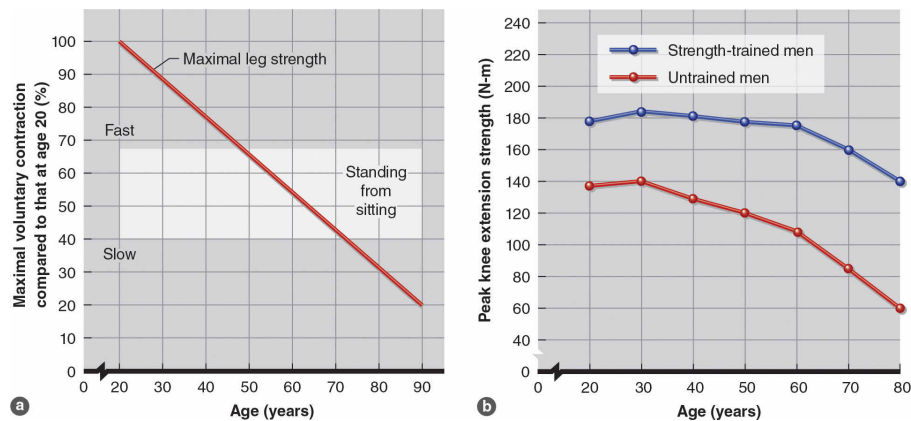


FIGURE 18.5 (a) The ability to stand from a sitting position is compromised at age 50, and by age 80 this task becomes impossible for some people. (b) Changes in peak knee extension strength in untrained and trained men at various ages. Note that older men (e.g., 60-80 years) who strength train can have knee extension strength equal to or greater than that of individuals who are only a third their age.

Age-related loss of muscle strength results primarily from the substantial loss of muscle mass that accompanies aging or decreased physical activity (or both), as discussed earlier in this chapter. Figure 18.6 shows a computed tomographic (CT) scan of the upper arms of three 57-year-old men of similar body weight (about 78-80 kg, or 172-176 lb). Note that the untrained subject has substantially less muscle and more fat than the others. The swim-trained subject has less fat and a markedly larger triceps muscle than the untrained subject, but his biceps muscle, a muscle seldom used during swimming, is not much different. However, both of these muscles are larger in the resistance-trained subject. The differences between these three men are likely attributable to a combination of genetics and their volume and type of training.

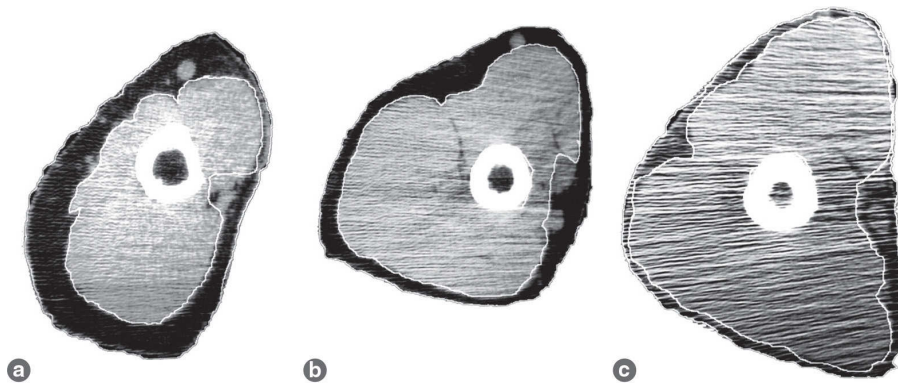


FIGURE 18.6 Computed tomography scans of the upper arms of three 57-year-old men of similar body weights. The scans show bone (dark center surrounded by white ring), muscle (striated gray area), and subcutaneous fat (dark perimeter). Note the difference in the muscle areas of (a) the untrained man, (b) the swim-trained man, and (c) the strength-trained man.

VIDEO 18.2 Presents Luc Von Loon on functional strength and the importance of strength training for both athlete and elderly populations.



Aging has a marked effect on muscle mass and strength, but what about muscle fiber type? There are conflicting results about the effects of aging on type I and type II fibers. Postmortem cross-sectional comparisons of the vastus lateralis (quadriceps) muscle in 15- to 83-year-old men and women have suggested that fiber type remains unchanged throughout life.²² However, results from longitudinal studies conducted over a 20-year period indicate that the frequency or intensity of activity, or perhaps both, might play an important role in fiber type distribution with aging.^{36,37} Muscle biopsy samples from the gastrocnemius (calf) muscles of a group of

previously elite distance runners, obtained in 1970 through 1974 and again in 1992, demonstrated that the runners who had decreased their activity (fitness trained) or become sedentary (untrained) had a significantly greater proportion of type I fibers than when they were 18 to 22 years younger ([figure 18.7](#)). Those who continued to train had no change. Although some of the elite runners who still competed in distance running (highly trained) showed a small increase in the percentage of type I fibers, on average these highly trained runners showed no change in their calf muscle fiber composition over the 18- to 22-year span of this study.

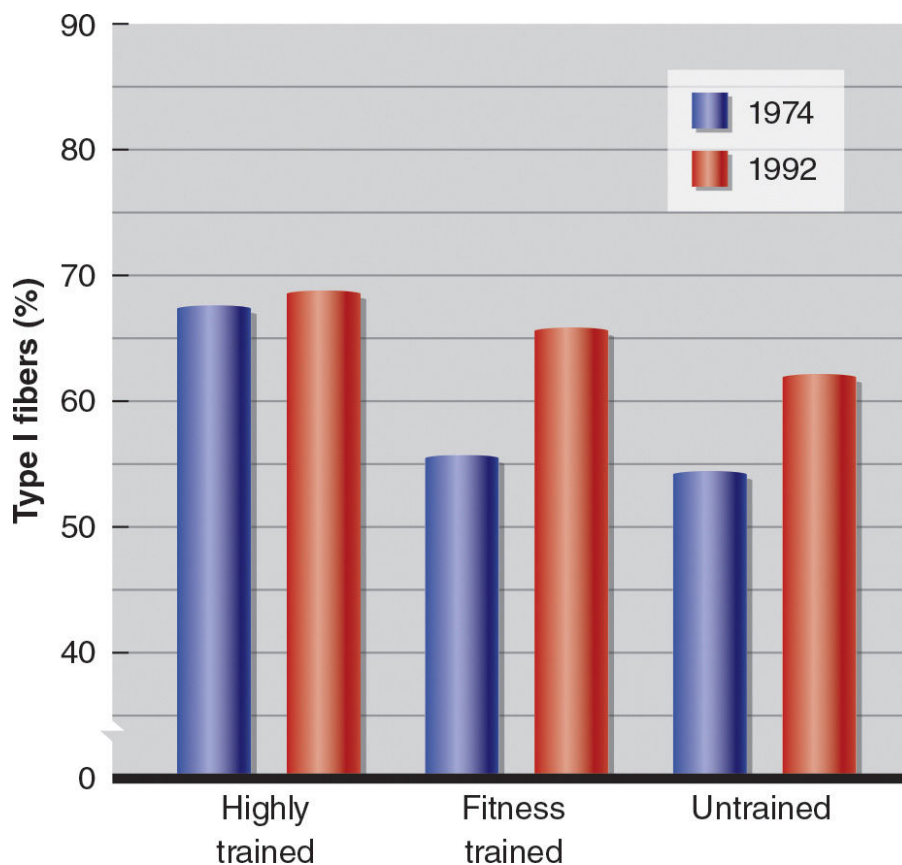


FIGURE 18.7 Changes in muscle fiber type composition of the gastrocnemius in elite distance runners who remained highly trained, stayed fitness trained, or became untrained during the 18 to 22 years between tests. Note that the runners who continued to compete showed almost no change in percentage of type I fibers, whereas the less fit and untrained individuals experienced an increase in the percentage of type I fibers.

It has been suggested that the apparent increase in type I fibers may be attributable to an actual decrease in the number of type II fibers, resulting in a greater *relative* proportion of type I fibers.

Although the precise cause of type II fiber loss is unclear, it has been suggested that the number of type II motor neurons decreases during aging, which eliminates innervation of these muscle fibers.

Numerous investigations have shown a decrease in both the number and the size of muscle fibers with aging. One study reported a loss of approximately 10% of the total number of muscle fibers per decade after age 50.²⁷ This explains in part the muscle atrophy that occurs as we get older. It appears that the size of both type I and type II fibers decreases with aging. Endurance training (such as distance running) has little impact on the decline in muscle mass with aging. Strength training, on the other hand, reduces muscle atrophy in aging adults and can, in fact, cause older individuals to increase their muscle cross-sectional area.²⁷

Aging is accompanied by substantial changes in the nervous system's capacity to process information and to activate muscles. Specifically, aging affects the ability to detect a stimulus and process the information to produce a response. Simple and complex movements are slowed with aging, although people who remain physically active are only slightly slower than younger, active individuals. Motor unit activation is lower in aged adults. For example, one study showed that older men (~80 years) had lower firing rates and longer twitch contraction durations, whereas younger men (~20 years) had relatively higher firing rates and shorter contraction times.⁴ However, others have shown that older individuals retain their ability to maximally recruit skeletal muscle, suggesting that reduced strength is due to local muscle rather than neural factors.

Neuromuscular changes during aging are at least partially responsible for decreased strength and endurance, but active participation in exercise and sport tends to lessen the impact of aging on performance. This doesn't mean that regular physical activity can arrest biological aging, but an active lifestyle can markedly reduce many of the decrements in physical work capacity.

Despite the loss of muscle mass in active aging men, the structural and biochemical properties of the remaining muscle mass are well maintained. The number of capillaries per unit area is similar in young and old endurance runners. Oxidative enzyme activities in

the muscles of endurance-trained older athletes are only 10% to 15% lower than in endurance-trained young athletes. Thus, oxidative capacity of skeletal muscle of endurance-trained older runners is only slightly less than that of young elite runners, which suggests that aging has little effect on skeletal muscle's adaptability to endurance training.

In Review

- Maximal strength decreases steadily with aging.
- Age-related losses of strength result primarily from a substantial loss of muscle mass.
- In general, normally active people experience a shift toward a higher percentage of type I muscle fibers as they age, possibly attributable to a reduction in type II fibers.
- The total number of muscle fibers and the fiber cross-sectional area decrease with age, but resistance training appears to lessen the decline in fiber area.
- Aging slows the nervous system's ability to respond to a stimulus, process the information, and produce a muscular contraction.
- Whereas endurance training does little to prevent the age-associated loss in muscle mass, strength training can maintain or increase the muscle fiber cross-sectional area in older men and women.

Cardiovascular and Respiratory Function

Changes in endurance capacity that accompany aging can be attributed to decrements in both central and peripheral cardiovascular function. Changes in respiratory function that accompany primary aging likely play a lesser role, although the incidence of respiratory dysfunction increases with age. In this section, we examine aging effects on both the cardiovascular and respiratory systems.

Cardiovascular Function

Similar to muscle function, cardiovascular function declines as we age. One of the most notable changes that accompanies aging is a decrease in maximal heart rate ($HR_{\max}$). $HR_{\max}$ is estimated to decrease slightly less than 1 beat/min per year as we age.

Traditionally, the average HR_{max} for any age has been estimated from the equation $HR_{max} = 220 - \text{age}$. This equation tends to overestimate the HR_{max} of children and young adults and underestimate the HR_{max} of older adults. Also, when the old equation ($HR_{max} = 220 - \text{age}$) is used, individual values can deviate by ± 20 beats/min from the predicted value. For example, the old equation predicts that an *average* 60-year-old would have an HR_{max} of 160 beats/min; however, an individual's actual HR_{max} might be as low as 140 beats/min or as high as 180 beats/min.

Tanaka and colleagues proposed a more accurate equation that seems to be appropriate for all people and is not influenced by sex or activity level.³⁵

$$HR_{max} = [208 - (0.7 \times \text{age})]$$

While this latter equation improves the prediction of the average individual's heart rate, there is still substantial interindividual variability. We will see in [chapter 20](#) that overestimates and underestimates make a big difference when HR_{max} is used for exercise prescription.

The reduction in HR_{max} with aging is not affected by training; that is, it appears to be similar for sedentary and well-trained adults. This reduction in HR_{max} might be attributable to morphological and electrophysiological alterations in the cardiac conduction system, specifically in the sinoatrial (SA) node and in the atrioventricular (AV) bundle, which could slow cardiac conduction. Downregulation of the β_1 -adrenergic receptors in the heart also decreases the heart's sensitivity to catecholamine stimulation.

The decrease in maximal cardiac output with aging in highly trained men and women is primarily attributable to decreased heart rate and, to a lesser extent, a decrease in stroke volume. Maximal stroke volume (SV_{max}) is modestly reduced ($\sim 10\%$ - 20% reduction) in highly trained older adults. The responses to catecholamine stimulation and myocardial contractility are reduced, and recent evidence provided with more sophisticated Doppler imaging techniques indicates that the Frank-Starling mechanism is less effective due to left ventricular and arterial stiffening. Studies of endurance runners have shown that the lower $\dot{V}O_{2max}$ values

observed in older athletes result from a reduction in maximal cardiac output, despite the fact that heart volumes of older athletes are similar to those of young athletes, confirming that a decreased maximal heart rate is the primary cause of reduced $\dot{V}O_{2\max}$. In untrained men and women, a number of studies have demonstrated a clear and larger decrease in maximal stroke volume—and maximal cardiac output—with aging.

VIDEO 18.3 Presents Ben Levine on the factors that affect $\dot{V}O_{2\max}$ and cause its decline with aging.



Peripheral blood flow decreases with aging, even though capillary density in the muscles is unchanged. In middle-aged athletes, there is a 10% to 15% reduction in blood flow to working muscles in the leg at any given work rate when compared with well-trained young athletes ([figure 18.8](#)). This attenuation in blood flow is due to a number of peripheral factors including blunted functional sympatholysis (i.e., a greater sympathetic vasoconstriction in the exercising muscle) and a reduction in local vasodilators and their effect. But the reduced blood flow to the legs of these middle-aged and older endurance runners during submaximal exercise is apparently compensated for by a greater arterial–mixed venous oxygen difference, or $(a-\bar{v})O_2$ difference (more oxygen extracted by the muscles). As a result, although the blood flow is lower, oxygen uptake by the exercising muscles is similar at a given submaximal work intensity in the older age group. This was confirmed in a study of endurance-trained men, which compared subjects aged 22 to 30

years with subjects aged 55 to 68 years. Leg blood flow, vascular conductance, and femoral venous oxygen saturation were each 20% to 30% lower in the older men at each submaximal work rate, while leg $(a-\bar{v})O_2$ difference was higher in older subjects.³¹

It is hard to determine the extent to which the age-related changes in stroke volume, cardiac output, and peripheral blood flow result from primary aging and the extent to which they are attributable to the **cardiovascular deconditioning** that accompanies reduced activity. These declines in cardiovascular function with aging are largely responsible for the declines observed in $\dot{V}O_{2\max}$, as discussed later in this chapter.

Additionally, sedentary aged humans have a decrease in cardiac and arterial compliance caused by stiffening of the heart and the large elastic arteries. The aged blood vessels also exhibit a change in local control of blood flow including a dysfunction in the ability of the endothelium to release and respond to vasodilators such as nitric oxide and prostaglandins, termed **endothelial dysfunction**. This change contributes to the inability of blood vessels to vasodilate and the reduction in peripheral muscle blood flow during exercise.

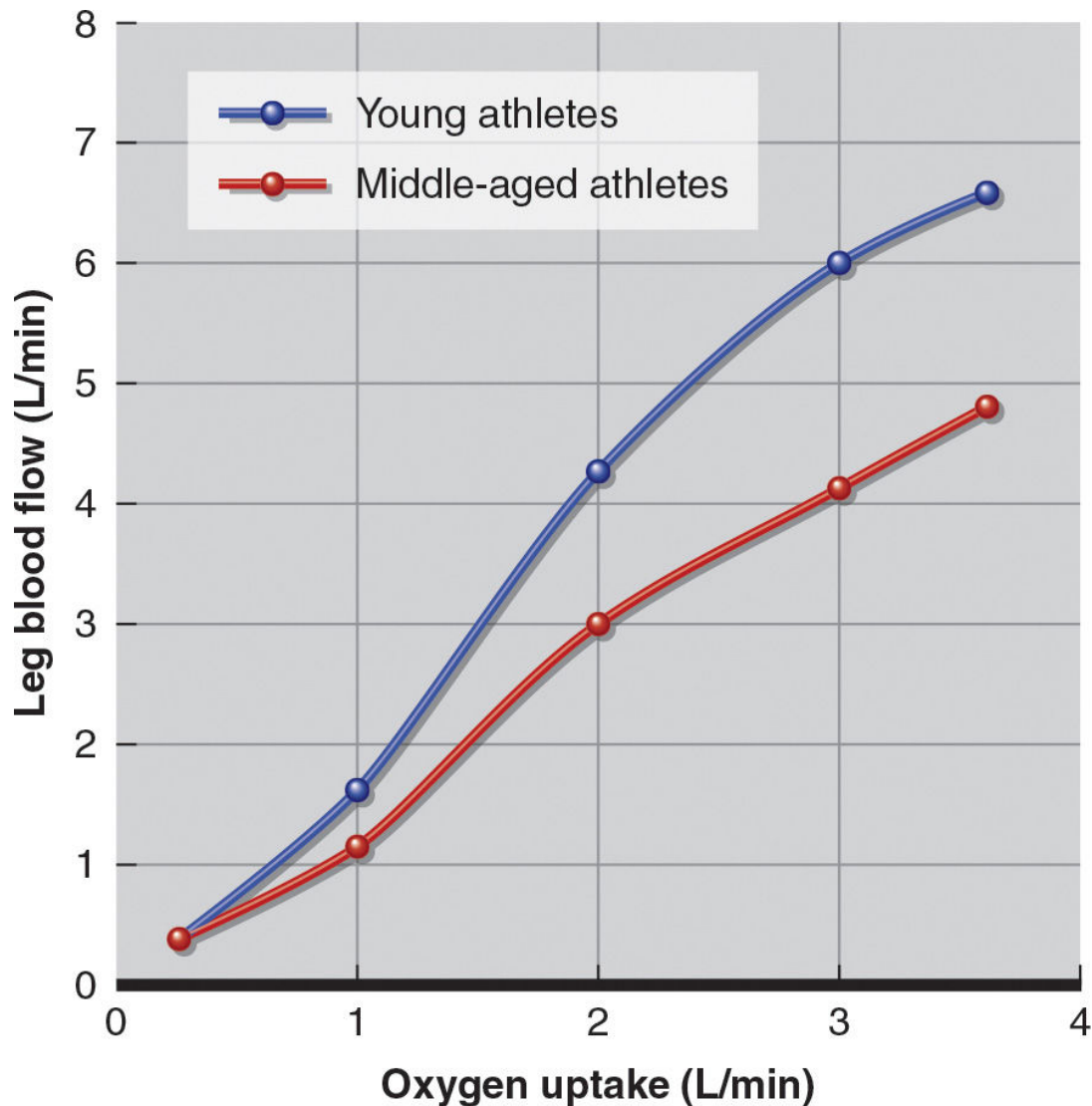


FIGURE 18.8 Leg blood flow during cycling exercise in young and middle-aged orienteers.

Adapted by permission from B. Saltin, *The Aging Endurance Athlete*, in *Sports Medicine for the Mature Athlete*, edited by J.R. Sutton and R.M. Brock (Indianapolis: Benchmark Press, 1986). Copyright 1986 Cooper Publishing Group, Carmel, IN.

Habitual exercise training in older men and women offsets some of the arterial stiffening and endothelial dysfunction. Part of the reason for this preservation in vascular function with habitual exercise training is a preservation or restoration in vasodilator signaling, including an increase in nitric oxide bioavailability. Research is ongoing to determine the appropriate dose of exercise (time, duration, and intensity) needed to see these positive cardiovascular benefits in healthy and clinical older populations.

Respiratory Function

Pulmonary function changes with aging, but these changes do not appear to limit exercise capacity. Both vital capacity (VC) and **forced expiratory volume in 1 s ($FEV_{1.0}$)** decrease linearly with age, starting at age 20 to 30. Whereas these decrease, residual volume (RV) increases, and the total lung capacity (TLC) remains essentially unchanged. As a result, the ratio of the residual volume to total lung capacity (RV/TLC) increases, meaning that less air can be exchanged. In our early 20s, RV accounts for 18% to 22% of the TLC, but this increases to 30% or more as we reach age 50. Smoking appears to accelerate this increase.

These changes are matched by changes in maximal ventilatory capacity during exhaustive exercise. **Maximal expiratory ventilation ($\dot{V}_{E\max}$)** increases during growth until people achieve physical maturity, and then it decreases with age. $\dot{V}_{E\max}$ values average about 40 L/min for 4- to 6-year-old boys, increase to 110 to 140 L/min for fully mature men, and then decrease to 70 to 90 L/min for 60- to 70-year-old men. Girls and women follow the same general pattern, although their absolute values are considerably lower at each age, primarily because of smaller body size. Recall from [chapter 7](#) that TLC is directly proportional to height, which is why men often have higher values than women.

The changes in pulmonary function as adults get older result from several factors. The most important of these is a loss of elasticity of the lung tissue and chest wall, which increases the work involved in breathing. The resulting stiffening of the chest wall appears to be responsible for most of the reduction in lung function. But despite all these changes, the lungs still have a remarkable reserve and maintain an adequate diffusion capacity to permit maximal exertion, and do not appear to limit exercise capacity.

Endurance-trained older athletes have only slightly decreased pulmonary ventilation capacities, so the decreased aerobic capacity in older athletes cannot be attributed to changes in pulmonary ventilation. During strenuous exercise, both normally active older people and athletes can maintain near-maximal arterial oxygen saturation. Changes in the lungs and the blood's oxygen-carrying

capacity are not responsible for the observed decrease in $\dot{V}O_{2\max}$ seen in aging athletes. Rather, the primary limitation is apparently linked to the cardiovascular changes described earlier. Aging decreases maximum heart rate, which lowers maximal cardiac output and blood flow to the exercising muscles. Submaximal $(a-\bar{v})O_2$ difference is maintained in older exercisers, suggesting that O_2 extraction is well preserved with aging.

In Review

- Much of the decline in endurance performance associated with aging can be attributed to decreased cardiovascular function.
- Maximum heart rate decreases about 1 beat/min per year as we age. The average $HR_{\max}$ for a given age can be estimated by the following equation: $HR_{\max} = [208 - (0.7 \times \text{age})]$.
- Maximal stroke volume is only slightly reduced in older athletes, but their cardiac output decreases with age primarily because of the reduction in $HR_{\max}$. In untrained people, maximal stroke volume decreases due to left ventricular and arterial stiffening with aging.
- Peripheral blood flow also decreases with age; in trained older athletes, however, this is offset by an increased $(a-\bar{v})O_2$ difference.
- The decrease in $\dot{V}O_{2\max}$ with aging and inactivity is largely explained by decreased $\dot{Q}_{\max}$ due to decreased $HR_{\max}$. Maximal stroke volume is modestly reduced ($SV_{\max}$), and maximal $(a-\bar{v})O_2$ difference typically does not change much with aging.
- The decrease in $HR_{\max}$ is attributable largely to decreases in the heart's intrinsic rate but also could be caused by decreases in sympathetic nervous system activity and alterations in the cardiac conduction system.
- The decrease in $\dot{V}O_{2\max}$ with aging is primarily a function of reduced blood flow to the active muscles, which is associated with the reduction in maximal cardiac output.
- Habitual exercise can partially reverse or prevent many of the detrimental vascular changes that occur with aging, including reducing arterial stiffness and improving endothelial function.
- It is unclear how much of the decrease in cardiovascular function with aging is attributable to aging alone and how much is attributable to deconditioning because of decreased activity. However, many studies indicate that these

changes are attenuated in older athletes who continue to train, which indicates that inactivity plays a substantial role.

- Both vital capacity and forced expiratory volume decrease linearly with age. Residual volume increases, so total lung capacity remains unchanged. This increases the RV/TLC ratio, meaning that less air can be exchanged in the lung with each breath.
 - Maximal expiratory ventilation also decreases with age.
 - Pulmonary changes that accompany aging are primarily caused by a loss of elasticity in the lung tissue and the chest wall. However, trained older athletes have only slightly decreased pulmonary ventilation capacity.
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Aerobic and Anaerobic Function

In investigating the effect of aging on aerobic and anaerobic function during exercise, the focus is on two key variables— $\dot{V}O_{2\max}$ and the lactate threshold.

$\dot{V}O_{2\max}$

To determine how $\dot{V}O_{2\max}$ changes with aging, there are several important issues to consider. First, one must decide how to express the $\dot{V}O_{2\max}$ values—in liters per minute (L/min) or in liters per minute per kilogram of body weight to adjust for changes in weight ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$). While $\dot{V}O_{2\max}$ expressed in liters per minute may decrease little over a 10- to 20-year period, when the same subjects' values are expressed relative to body weight, there may be a large decrease if the individuals gain weight over the interval between the initial test and the final test. For nonweight-bearing exercise, such as cycling, using liters per minute is usually most appropriate. For weight-bearing activities such as running, it is usually more appropriate to express the values per unit of body weight ($\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$).

A second issue relates to whether change values in variables with aging should be expressed as an absolute change (L/min or $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) or as a percentage change.

This might seem like a minor point, but it is not. As an example, a 30-year-old man had an initial $\dot{V}O_{2\max}$ of $50 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, and at the age of 50 years his $\dot{V}O_{2\max}$ decreased to $40 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. A

60-year-old man had an initial $\dot{V}O_{2\max}$ of $35 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, and at the age of 80 years his $\dot{V}O_{2\max}$ decreased to $25 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$. In this example, both men have decreased their $\dot{V}O_{2\max}$ by $10 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ over a 20-year period, showing a decline of $0.5 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year. However, the younger man has had a decrease of 20% ($10/50 = 0.20$) over 20 years, or 1% per year, whereas the older man has had a decrease of 29% ($10/35 = 0.29$), or 1.4% per year. Although the two men have identical decreases in $\dot{V}O_{2\max}$ when expressed in $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, the older man has a substantially greater decrease when expressed as a percentage decrease. Keeping this in mind, let's consider changes in $\dot{V}O_{2\max}$ with aging, looking first at changes in normally active people and then at changes in highly trained endurance athletes.

Normally Active People The first studies of aging and physical fitness were performed by Sid Robinson³² in the late 1930s. He demonstrated that $\dot{V}O_{2\max}$ in normally active men declined steadily from ages 25 to 75 (table 18.1). His cross-sectional data show that aerobic capacity declines an average of $0.44 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year up to age 75, which is about 1% per year or 10% per decade. A review of 11 cross-sectional studies on men, most under age 70, showed that the average rate of decrease in $\dot{V}O_{2\max}$ was $0.41 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year for men and $0.30 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year for women, again close to 1% per year.¹

In the mid-1990s, a large cross-sectional study of changes in $\dot{V}O_{2\max}$ with aging was conducted at the NASA/Johnson Space Center in Houston, Texas. This study included 1,499 men and 409 women, all of whom were healthy and had performed a maximal treadmill test to exhaustion during which $\dot{V}O_{2\max}$ was directly measured.^{19,20} The authors reported a decline in $\dot{V}O_{2\max}$ of $0.46 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in men (1.2% per year) and $0.54 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in women (1.7% per year).

Unfortunately, few longitudinal studies have been conducted in this area. Nevertheless, the rate of decline in $\dot{V}O_{2\max}$ generally is agreed to be approximately 10% per decade or 1% per year ($-0.4 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year) in relatively sedentary men. The results are similar for women, although fewer subjects have been studied.

TABLE 18.1 Changes in $\dot{V}O_{2\max}$ Among Normally Active Men

Age (years)	$\dot{V}O_{2\max}$ (ml · kg ⁻¹ · min ⁻¹)	% change from 25 years
25	47.7	
35	43.1	-10
45	39.5	-17
52	38.4	-20
63	34.5	-28
75	25.5	-47

Data from Robinson (1938).

Older Athletes One of the most notable long-term studies of distance runners and aging was conducted by D.B. Dill and his colleagues from the Harvard Fatigue Laboratory.⁶ Don Lash, world record holder for the 2 mi run (8 min 58 s) in 1936, was among the athletes studied by the Harvard group. Although few of the former runners continued to train after leaving college, Lash was still running about 45 min per day at age 49. Despite this activity, his $\dot{V}O_{2\max}$ had declined from 81.4 ml · kg⁻¹ · min⁻¹ at age 24 to 54.4 ml · kg⁻¹ · min⁻¹ at age 49, a 33% decline. Runners who did not continue to train during middle age showed declines of about 43% from ages 23 to 50 (from 70 to 40 ml · kg⁻¹ · min⁻¹). These data suggest that prior training offers little advantage to endurance capacity in later life unless a person continues to engage in some form of vigorous activity. However, due to their high initial values, these individuals have a large functional reserve, and this large decrease in aerobic capacity has little effect on their ability to carry out activities of daily living. In addition, there are large individual differences in the rate of decline in $\dot{V}O_{2\max}$ with aging, and genetics is a major contributor.

More recent longitudinal studies of older male runners and rowers have shown a decline in aerobic capacity and cardiovascular function and changes in muscle fiber composition with aging. These athletes were studied for 20 to 28 years, during which time those athletes who continued high-volume and -intensity training experienced a 5% to 6% decline in $\dot{V}O_{2\max}$ per decade. On the other hand, elite runners who stopped training experienced nearly a 15% decline in aerobic capacity per decade (1.5% per year).

Fewer studies have been published on women, but the results show the same trends. In one study of 86 male and 49 female

masters endurance runners, the authors observed both cross-sectional and longitudinal (approximately 8.5 years) changes in $\dot{V}O_{2\max}$ with age.¹⁵ Their results are illustrated in figure 18.9. The average rate of decline, as indicated by the cross-sectional data regression line, was $0.47 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in men (0.8% per year) and $0.44 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ in women (0.9% per year). However, this figure shows that the longitudinal changes are greater than the cross-sectional changes, particularly for the older ages. In a cross-sectional study of sedentary women ($n = 2,256$), active women ($n = 1,717$), and endurance-trained women ($n = 911$) aged 18 to 89 years, $\dot{V}O_{2\max}$ declined by $0.35 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in the sedentary women (1.2% per year), $0.44 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in the active women (1.1% per year), and $0.62 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year in the endurance-trained women (1.2%).⁸

In the early 2000s, a 25-year follow-up study reexamined highly competitive, older male distance runners.^{37,38} These men were initially tested at 18 to 25 years of age. During the interval between testing sessions, the runners trained at about the same relative intensity as they had when they were younger. As a consequence, their $\dot{V}O_{2\max}$ values (L/min) declined only 3.6% over the 25-year period,³⁸ as shown in table 18.2. Although their maximal oxygen uptake decreased from 69.0 to $64.3 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, this is a decrease of only $0.19 \text{ ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ per year or 0.3% per year, and most of that change was attributable to a 2.1 kg (4.6 lb) increase in body weight.

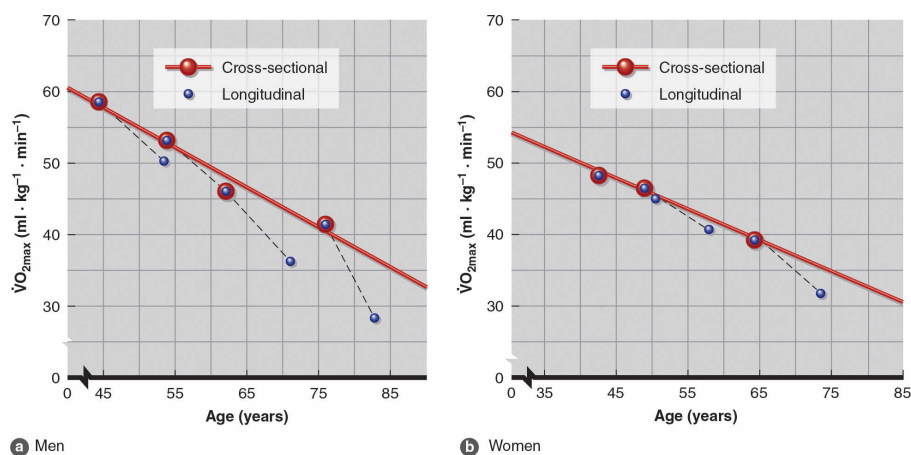


FIGURE 18.9 Cross-sectional and longitudinal declines in $\dot{V}O_{2\max}$ with age in a group of (a) 86 male and (b) 49 female masters endurance athletes.
Adapted from Hawkins et al. (2001).

This rate of decrease in these older runners' $\dot{V}O_{2\max}$ values is significantly less than that of either sedentary people or those who fitness train at levels and intensities below those of these older runners. Scott Trappe and colleagues³⁹ had the unique opportunity to examine aerobic capacity in two cohorts of healthy, independently living men aged 80 to 91 years who differed dramatically in their lifelong physical activity habits. The authors recruited a group of nine elite Swedish masters cross-country skiers who were still actively engaged in the sport. These octogenarian athletes were lifelong exercisers who had never stopped training for more than 6 months at any point in the past 50 years! A second subject sample involved six healthy age-matched controls who had no history of regular exercise. Each subject performed a $\dot{V}O_{2\max}$ test on a cycle ergometer and had a resting vastus lateralis muscle biopsy performed to measure such markers of aerobic capacity as the oxidative enzyme citrate synthase and the mitochondrial biogenesis initiator PGC-1 α .

TABLE 18.2 Changes in Mean Aerobic Capacity and Maximal Heart Rate with Aging in a Group of 10 Highly Trained Masters Distance Runners

$\dot{V}O_{2\max}$				
Age (years)	Weight (kg)	(L/min)	(ml · kg ⁻¹ · min ⁻¹)	HR _{max} (beats/min)
21.3	63.9	4.41	69.0	189
46.3	66.0	4.25	64.3	180

Not surprisingly, the lifelong endurance athletes exhibited a higher $\dot{V}O_{2\max}$, whether expressed in absolute (2.6 versus 1.6 L/min) or relative (38 versus 21 ml · kg⁻¹ · min⁻¹) terms. Skeletal muscle citrate synthase was 54% higher, and basal PGC-1 α was 135% higher in the athletes. Both mitochondrial enzyme activity and PGC-1 α were correlated with $\dot{V}O_{2\max}$. The combined cardiorespiratory and muscle

adaptations resulted in almost a 40% greater maximal power output on the cycle ergometer in the athletes. When one considers that some absolute $\dot{V}O_{2\max}$ is associated with simply being able to complete activities of daily living (see [figure 18.10](#)), the authors of this study clearly show the expanded functional reserve between this minimal fitness level and the uniquely high aerobic capacity of the octogenarian athletes that reflects a lower risk for disability and mortality. As the authors point out, not only is the 11-MET (metabolic equivalent) aerobic capacity of these aged athletes the highest ever recorded in this age group, “it places them in the lowest all-cause mortality risk category for men *of any age*”.

Although the number of studies on women is much smaller, a similar trend would be expected. Note that although intense training reduces the normal aging-related decline in $\dot{V}O_{2\max}$, aerobic capacity still declines. Thus, it appears that highly intense training has a slowing effect on the rate of loss in aerobic capacity during the early and middle years of adult life (e.g., 30-50 years) but less effect after 50 years of age.

In summary, $\dot{V}O_{2\max}$ declines with age, and the rate of decline is approximately 1% per year. Many factors influence this rate of decline, including the following:

- Genetics
- General activity level
- Intensity of training
- Volume of training
- Increased body weight and body fat mass, decreased fat-free mass
- Age range, with older individuals experiencing greater declines

There is not universal agreement on which of these factors are most important. An integrated view of the concepts of the physiological mechanisms contributing to the reductions in endurance performance with aging is presented in [figure 18.11](#).

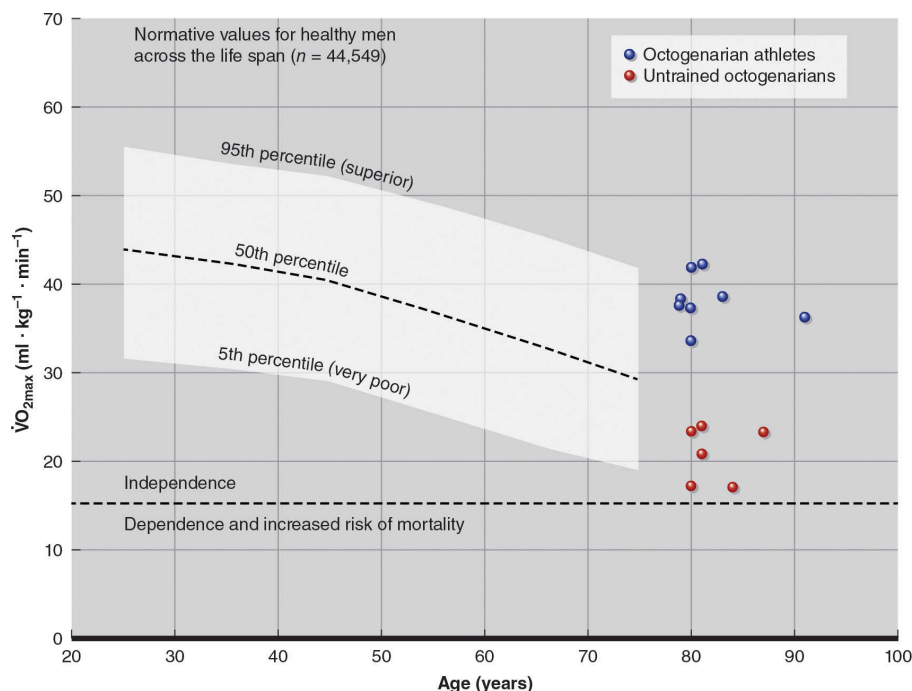


FIGURE 18.10 The figure shows individual $\dot{V}O_{2max}$ data from the lifelong endurance athletes and healthy untrained octogenarians studied by Trappe and colleagues. The dotted line at the bottom represents a 5-MET exercise capacity thought to be necessary for independently performing activities of daily living. The normative values for healthy men across the life span, shown in the shaded area, are derived from data from the Cooper Institute in Dallas, Texas.

Reprinted by permission from S. Trappe et al., "New Records in Aerobic Power Among Octogenarian Lifelong Endurance Athletes," *Journal of Applied Physiology* 114 (2013): 3-10.

Lactate Threshold

In young endurance-trained adults, the lactate threshold predicts exercise performance in distance events ranging from 3.2 km (2 mi) to the marathon. Lactate threshold expressed as a percentage of $\dot{V}O_{2max}$ (LT-% $\dot{V}O_{2max}$) provides the best marker relative to endurance running performance in individuals with similar $\dot{V}O_{2max}$ values. Few studies have addressed the changes in either the lactate threshold or the anaerobic threshold, derived from ventilatory variables, with aging. Lactate threshold was determined in a cross-sectional study of a group of masters endurance runners, 40 to 70+ years of age (111 men and 57 women).⁴⁰ Interestingly, LT-% $\dot{V}O_{2max}$ did not differ between men and women, but it did increase with age. More recent longitudinal studies with masters athletes reported that the change in lactate threshold over a 6-year follow-up was not predictive of running performance when it was expressed as a percentage of $\dot{V}$

$\dot{V}O_{2\max}$.²⁸ Another study reported similar results in 152 untrained men and 146 untrained women.²⁹ However, in both studies, $\dot{V}O_{2\max}$ was lower in the older groups, which helps explain the increase in LT-% $\dot{V}O_{2\max}$. When the LT at the absolute $\dot{V}O_2$ is compared between age groups, LT declined with age.

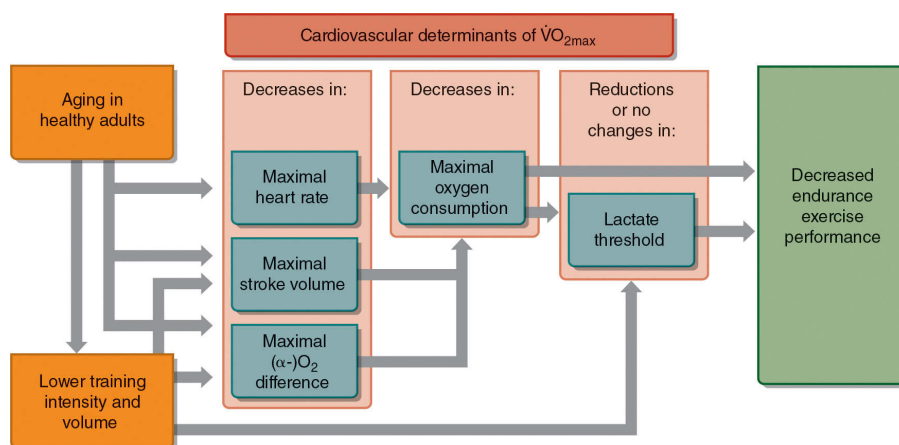


FIGURE 18.11 Factors and physiological mechanisms contributing to the reductions in endurance performance with advancing age in healthy humans. Primary aging directly contributes to the reduction in the cardiovascular determinants of $\dot{V}O_{2\max}$; however, there tends to also be a decrease in training volume and intensity with aging (dotted lines), so the relative contributions of each to reduced endurance performance are difficult to determine.



RESEARCH PERSPECTIVE 18.1

Centenarian Athletes

Centenarians (people who live to or beyond the age of 100 years) are often considered models of healthy aging. Many reach this age while maintaining generally good health, and some of them preserve a remarkably high level of physical activity. Indeed, elite athletes live longer than the general population.³ Although these individuals may be associated with having outstanding performance capacity, the age-related decline in performance in centenarians was only recently investigated.

A group of researchers identified and compared all of the best performances achieved by centenarians in athletics, swimming, and yoga.²⁶

Within each discipline, the decline in performance was expressed as a percentage of the world record for that discipline. The researchers noted that the average decline in performance of centenarian athletes was 78% in comparison with the initial world record, consistent with the concept that performance declines by ~10% during each decade of life after the age of 35 years. The best performance achieved by a centenarian was in track cycling, potentially owing to improvements in skeletal muscle cycling efficiency that offset limitations in oxygen transport. It is thought that as the number of centenarians continues to grow and as attitudes about exercise change among older adults, these types of exceptional performances by athletes will become more common.

In Review

- It is often difficult to differentiate between the results of biological aging and physical inactivity. A natural deterioration in physiological function occurs with aging, but this is compounded by the fact that most people also become more sedentary as they age.
 - Aerobic capacity generally decreases by about 10% per decade or 1% per year in relatively sedentary men and women.
 - If highly trained endurance athletes begin with a higher $\dot{V}O_{2\max}$ and $\dot{V}O_{2\max}$ declines at the same rate, it will remain higher than that of sedentary people of the same age.
 - Studies of older athletes and less active people of the same age group indicate that the decrease in $\dot{V}O_{2\max}$ is not strictly a function of age. Athletes who continue to train have significantly smaller decreases $\dot{V}O_{2\max}$ as they age, particularly if they train at a high intensity.
 - Lactate threshold expressed as a percentage of $\dot{V}O_{2\max}$ increases with aging, but it decreases when expressed in relation to the absolute $\dot{V}O_2$ at which it occurs.
-

Physiological Adaptations to Exercise Training

Despite the decrements in body composition, $\dot{V}O_{2\max}$, and strength associated with aging, well-trained middle-aged and older athletes are capable of exceptional performances. Furthermore, those who train for general fitness appear to experience gains in muscular strength and endurance similar to those of young adults.

Strength

As with most other physiological functions, the loss of strength with aging is likely the result of a combination of primary aging and reduced physical activity that produces a decline in muscle mass and function. Although it is difficult to compare the adaptations to resistance training of younger and older people, aging appears to neither impair the ability to improve muscle strength nor prevent muscle hypertrophy. For example, when older men (aged 60-72) resistance trained for 12 weeks at 80% of their 1-repetition maximum (1RM) for extension and flexion of both knees, their extension strength increased by 107% and flexion strength increased by 227%.⁹ These improvements were attributed to muscle hypertrophy as determined from mid thigh CT scans. Biopsies of the vastus lateralis muscle (in the quadriceps) revealed that the cross-sectional area of type I fibers increased by 34% and that of type II fibers by 28%. The greater increase in strength compared to size was due to relatively low baseline strength values and likely neural adaptations in these previously sedentary men. In another study of older untrained men (aged 64 years), a 16-week resistance training program resulted in major increases in strength (50% for leg extension strength, 72% for leg press strength, and 83% for half-squat strength) and an increase in the mean cross-sectional area of all major muscle fiber types (46% for type I, 34% for type IIx, and 52% for type IIb).^{12,16}

In a study of older women (average age 64 years), 21 weeks of resistance training resulted in a 37% increase in leg extensor maximal force development; a 29% increase in leg extension 1RM; an increase in the cross-sectional area of the extensor muscles; and a 22% to 36% increase in type I, type IIx, and type IIb muscle fiber areas.¹³

In general, older individuals can experience a significant benefit from resistance training. Older resistance-trained athletes tend to have higher muscle mass, are generally leaner, and are approximately 30% to 50% stronger than their sedentary peers. Furthermore, compared to age-matched aerobic exercise-trained subjects, resistance-trained subjects have more total muscle mass, have higher bone mineral densities, and maintain higher muscle

strength and power. Although not to the same extent as older athletes, older sedentary individuals also experience significant strength gains from resistance training, which greatly improves their abilities to perform activities of daily living and helps to prevent falling.

Neuromuscular Function

As discussed previously, the process of aging induces clear changes in muscle, but exercise training can induce specific adaptation to slow the progression of some of these age-related changes. Because neuromuscular adaptations to exercise training are some of the first changes to occur, the impact of training on the neuromuscular junction (NMJ) is of particular interest.

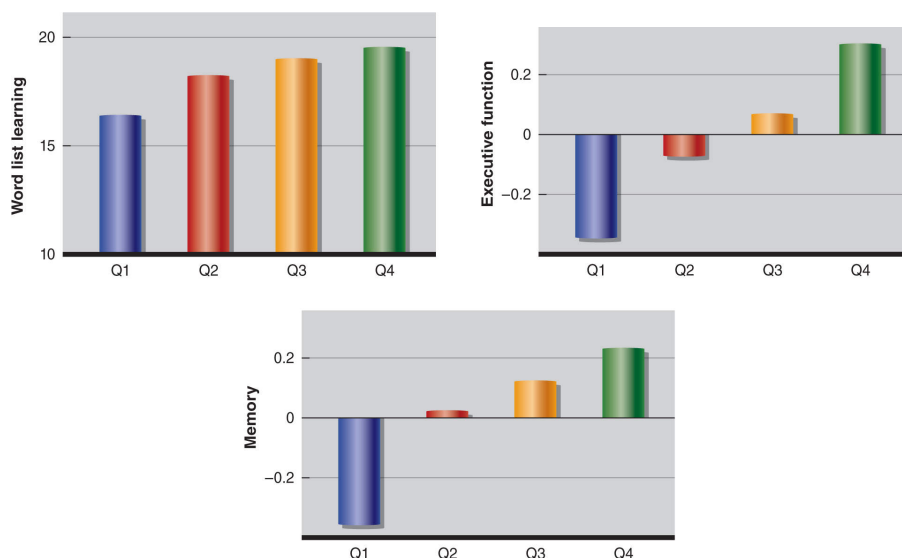
To examine how aging affects the adaptations of the NMJ to exercise training, groups of old and young rats were trained on a treadmill while sedentary rats served as controls.⁵ After the training program, slow- and fast-twitch fibers from different muscles were imaged to look at adaptations at the pre- and postsynaptic end plates using high-powered confocal microscopy, a technique that creates high-resolution three-dimensional images of microscopic structures. Additionally, the cross-sectional area of the muscle fibers was measured to look at differences in the size of the muscle fibers themselves. Neuromuscular junctions in slow-twitch muscle fibers in the young rats demonstrated significant adaptations to exercise training, but those in the old rats showed no change. On the other hand, NMJs in many fast twitch-fibers were affected by age, but training did not induce any beneficial adaptations in the NMJs. Changes in muscle fiber size were unrelated to changes in NMJs. Taken together, these data suggest that aging interferes with the ability of the NMJ to adapt to exercise training and show how complex the interactions of different muscles and fiber types are in their response to both aging and training.

RESEARCH PERSPECTIVE 18.2

Physical Activity and Cognitive Function in Older Adults

The population continues to age, and as it does, cognitive impairment and other forms of dementia are becoming a growing health threat. Physical activity reduces the risk of chronic disease, and studies support the concept that physical activity also delays the onset of cognitive decline and the incidence of dementia or Alzheimer's disease associated with aging. However, most of these studies have relied on self-report measures of physical activity, which have many inherent limitations. Thus, more objective assessments of physical activity are required to accurately define the dose–response relation of physical activity with cognitive function.

The REasons for Geographic and Racial Differences in Stroke (REGARDS) study is a national longitudinal study of ~6,500 black and white older adults that investigated the causes of regional and racial disparities in stroke mortality. As a part of this study, researchers investigated the association of objectively measured physical activity (via accelerometry) with longitudinal cognitive function (global cognitive function, memory, and executive function) in older adults.⁴¹ The findings demonstrate a significant relation between physical activity and cognitive function in both white and black older adults (see figure). Moreover, at follow-up (2 years after study entry), those with a greater percentage of moderate- to vigorous-intensity physical activity demonstrated less of a decline in cognitive function. Even relatively minor differences in the percentage of time spent engaging in moderate- to vigorous-intensity physical activity (i.e., 3-5 min/day) was associated with a 36% lower risk of incident cognitive impairment and greater maintenance of executive function and memory over time. These relations were evident in both white and black older adults. Collectively, these findings demonstrate a dose–response relation between physical activity and cognitive performance over time. Thus, intervention strategies aimed at preventing declines in cognitive function should be initiated before symptoms manifest.



Physical activity and cognitive function. Cognitive scores, presented for an index of cognitive function, executive function, and memory, in older adults. Participants (~70 years) were grouped into quartiles based on the percentage of time spent engaging in moderate- to vigorous-intensity physical activity via accelerometry (Q1: 0-0.17%; Q2: 0.17-0.75%; Q3: 0.75-2.14%; Q4: $\geq 2.14\%$). Word List Learning (WLL) is a test to identify early cognitive decline and is from the Consortium to Establish a Registry for Alzheimer's Disease battery of tests. Physical activity is associated with improvements in cognitive function, executive function, and memory in a dose-dependent manner.

Age-related changes in the motor unit have a profound effect on motor function, especially among the expanding number of older adults. A recent review¹⁸ presented evidence that age-related changes in motor unit morphology and properties lead to impaired motor performance. The resulting changes included reduced maximal strength and power, slower contractile velocity, and increased fatigability. Additionally, older subjects showed increased variability while performing motor tasks due to reduced and more variable synaptic inputs that drive motor neuron activation, fewer and larger motor units, less stable neuromuscular junctions, lower and more variable motor unit action potential discharge rates, and smaller and slower skeletal muscle fibers. The impact of regular physical activity on such variability in motor performance is unknown.

Aerobic and Anaerobic Capacity

Studies have shown that improvements in $\dot{V}O_{2\max}$ with training are similar for younger (ages 21-25) and older (ages 60-71) men and women.^{24,29} Although the pretraining $\dot{V}O_{2\max}$ values were, on the average, lower for the older subjects, the absolute increases of 5.5 to 6.0 ml $\cdot$ kg⁻¹ $\cdot$ min⁻¹ were similar in the two groups. Additionally, older men and women experienced similar increases in $\dot{V}O_{2\max}$, averaging 21% for men and 19% for women, when they trained for 9 to 12 months by walking, running, or both walking and running about 4 mi (6 km) per day. Cardiovascular adaptation to training in previously sedentary older subjects appear to peak after 3 to 6 months of moderate training.³³ Together, these studies indicate that endurance training produces similar gains in aerobic capacity in healthy people throughout the age range of 20 to 70 years, and this adaptation is independent of age, sex, and initial fitness level.

However, this does not mean that endurance training can enable older athletes to achieve the performance standards established by younger athletes.

The precise mechanisms that trigger the body's adaptations to training at any age are not fully understood, so we do not know if improvements from training are achieved in the same way throughout life. For example, much of the improvement in $\dot{V}O_{2\max}$ seen in younger subjects is associated with an increase in maximal cardiac output. But older subjects show significantly greater gains in muscle oxidative enzyme activities, which suggests that peripheral factors in older subjects' muscles might play a greater role in aerobic adaptations to training than in younger subjects.

Very little is known about the trainability of anaerobic capacity in older people. We saw earlier in this chapter that lactate threshold, expressed as a percentage of a person's $\dot{V}O_{2\max}$, increases with aging and is not associated with endurance running performance. In young and middle-aged adults, LT-% $\dot{V}O_{2\max}$ is the best predictor of endurance performance—running, cycling, swimming, and cross-country skiing. As we have already seen, different rates of aging of the oxygen transport and lactic acid buffering systems are a likely explanation for this difference between older adults and young and middle-aged adults. A related point is that when one is comparing the lactate threshold of individuals with different $\dot{V}O_{2\max}$ values, it is likely more appropriate to consider the absolute $\dot{V}O_2$ value to which that threshold corresponds when attempting to explain endurance performance.

Mobility

One of the major problems associated with aging is a loss of mobility, the ability to move freely and with minimal restrictions or pain. Loss of mobility is associated with increased disability, hospitalization, and mortality. The Lifestyle Interventions and Independence for Elders (LIFE) study sought to determine whether a structured physical activity program could reduce the risk of major mobility problems in a cohort of sedentary men and women aged 70 to 89 years.³⁰

A sample of 1,635 sedentary men and women who had physical limitations but were able to walk 400 m were recruited, then randomized to either a structured, moderate-intensity physical activity program that included aerobic, resistance, and flexibility training, or a control group. After 2.6 years, they tested both groups' ability to walk 400 m, which they defined as a major mobility disability. Mobility disability occurred in 36% of the control group but only 30% of the physical activity group. The authors concluded that a structured, moderate-intensity physical activity program can reduce major mobility disability over 2.6 years among older adults at risk for disability.

Another recent study also showed improvement in mobility after 20 weeks of resistance training in individuals over the age of 65.¹⁰ Interestingly, training older men and women with fast, unloaded movements improved mobility as well as conventional resistance training does. The results were attributed to neural adaptations that activate more type II muscle fibers. Older people do not have to lift heavy weights or use specialized equipment to achieve neuromuscular benefits that improve mobility—similar adaptations occurred when older subjects were trained to move their limbs as quickly as possible.

RESEARCH PERSPECTIVE 18.3

Age-Related Changes in Human Skeletal Muscle

The term sarcopenia, based on the Greek words *sarx* (flesh) and *penia* (poverty), defines the age-related loss of muscle mass. In addition to loss of muscle mass, there are also notable changes in muscle architecture and molecular structure and function in older adults, all of which ultimately manifest as limitations in muscle function. Reductions in muscle force production (strength) and endurance are features of sarcopenia and can be attributed to decreases in muscle fiber number as well as fiber cross-sectional area. Recently, mitochondria have been implicated as potential contributors to sarcopenia.²

Mitochondria are the main suppliers of cellular energy but are also primary contributors to the generation of reactive oxygen species (ROS). It appears that mitochondrial function, in general, is impaired in aging, including reductions in mitochondrial protein synthesis, respiration, and maximal rate of ATP production, partly as a result of increased uncoupling of oxygen

consumption to ATP synthesis. Moreover, recent studies have suggested that mitochondrial dysfunction may contribute to muscle atrophy. This linkage to decrements in muscle performance indicates that altered mitochondrial function and morphology may promote intracellular oxidative stress, which can interfere with myofilament function via oxidative modification of proteins or disruptions in myofilament protein metabolism. Therefore, the relation between alterations in mitochondrial biology and deficits in muscle and myofilament structure and function likely contribute to the age-related impairments in skeletal muscle function. Interestingly, it appears that this relation may be modulated by aerobic exercise and habitual physical activity.

It is well known that exercise is a powerful stimulus for mitochondrial biogenesis in young adults. In sedentary older adults, a decline in mitochondrial content and function is evident at the mRNA, protein, morphological, and function (e.g., respiration) levels. In rodent models of aging, as well in humans, part of this loss of function can be alleviated by exercise training, suggesting that low physical activity likely contributes to mitochondrial dysfunction. In older adults, exercise increases mitochondrial content and provides a greater capacity to defend against oxidative stress. Habitual exercise has also been demonstrated to improve mitochondrial respiration in older adults. Collectively, it appears that exercise can be used as an effective intervention strategy to improve mitochondrial dysfunction in aged muscle, thereby restoring coupling efficiency and improving overall muscle function and performance.

In Review

- Older adults appear to get the same benefits—maintenance of body weight, decreased percent body fat and fat mass, and increased fat-free mass—from exercise training as younger and middle-aged adults.
- It appears that aging does not impair a person's ability to increase muscle strength or muscle hypertrophy. Individual muscle fibers of older individuals also have the ability to increase in size.
- Endurance exercise training produces similar absolute gains in healthy people, regardless of their age, sex, or initial level of fitness. However, percent improvement is larger in those with low initial baseline levels.
- With endurance training, an increased $\dot{V}O_{2\max}$ in older exercisers results mostly from improvement in the muscles' oxidative enzyme activities (peripheral adaptation), whereas improvement in younger people is largely attributable to increased maximal cardiac output (central adaptation).
- Age-related changes occur in motor unit structure and function that lead to impaired motor performance.

- Loss of mobility with aging can be offset by a variety of exercise programs and do not necessarily require lifting heavy weights.
-

Sport Performance

World and national records in running, swimming, cycling, and weightlifting suggest that people are in their physical prime during their 20s or early 30s. If we use a cross-sectional approach, comparing these records with national and world records for older athletes in these events allows us to examine the effects aging has on the best performers. Unfortunately, we have little longitudinal information about the effects of aging on performance because few studies have enabled us to follow physical performance in selected individuals over the span of their athletic careers. However, we can look historically at performance times in certain athletic events to gain insight into the influence of physiological function on performance with aging. In the following sections, we consider how aging affects certain types of sport performance.

Running Performance

In 1954, Roger Bannister, a 21-year-old medical student, stunned the sporting world when he became the first person to run the mile (1.61 km) in less than 4 min (3 min 59.4 s). The current record for the mile is 3:43:13, set by Hicham El Guerrouj of Morocco in 1999, which is more than 16 s faster than Bannister's record—a gap that would have placed Bannister more than 100 m (109 yd) behind today's record holder. In 1954, it seemed inconceivable that a sub-4 min mile could have been accomplished by someone over the age of 30 years. The oldest individual to record a sub-4 min mile was Eamonn Coghlan, who was 41 years old when he ran it indoors in 3:58:13. The oldest individual to record a sub-5 min mile was 65 years old.

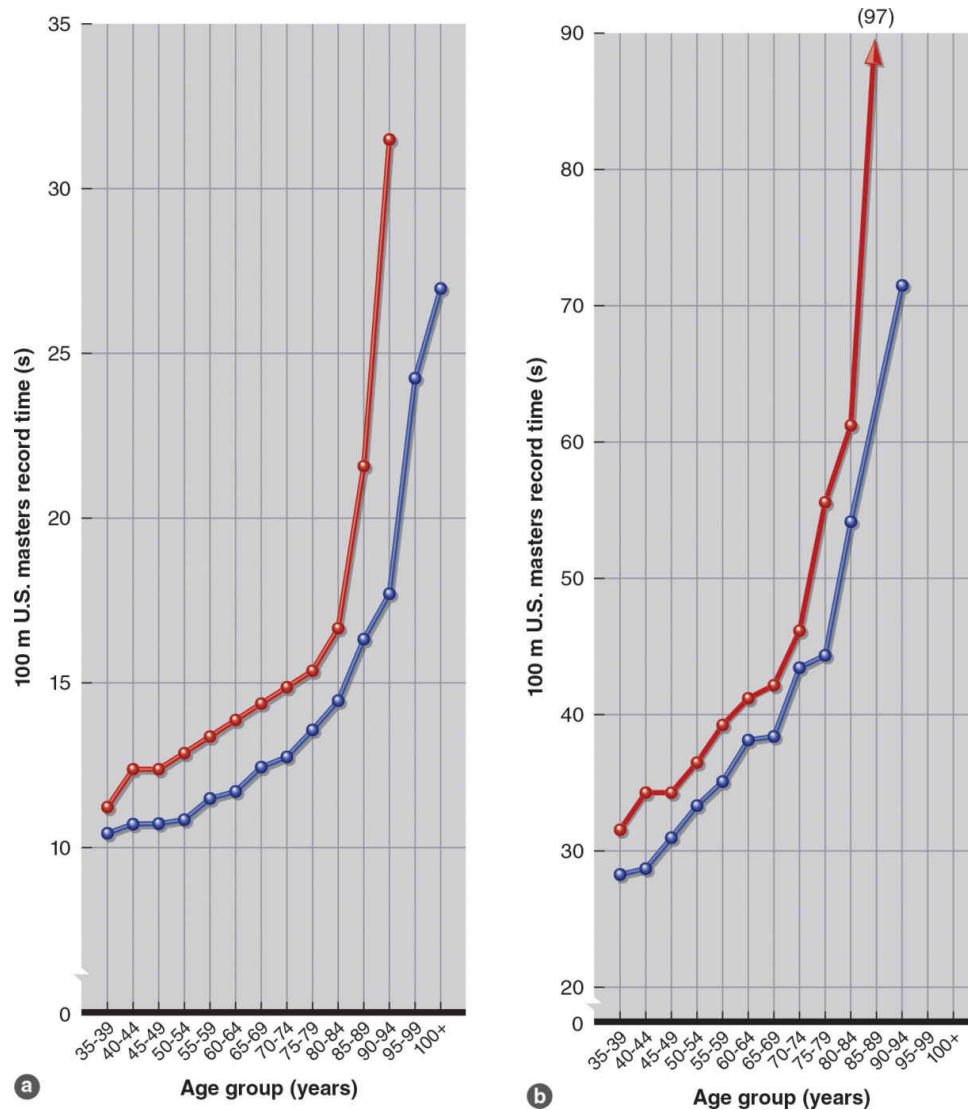


FIGURE 18.12 Men's and women's U.S. masters records across age groups for (a) 100 m and (b) 10 km runs.

Although older runners have achieved some exceptional records, running performance in general declines with age, and the rate of this decline appears to be relatively independent of distance. Longitudinal studies of elite distance runners indicate that despite a high level of training, performance in events from the mile (1.61 km) to the marathon (42 km) declines at a rate of about 1% per year from the age of 27 to 47 years.^{37,38} It is interesting to note that age-group masters records for both 100 m and 10 km runs decrease fairly linearly from age 25 to age 70 or so, as shown in [figure 18.12](#). Beyond age 75 to 80, however, the records for men and women drop

more precipitously. A sprint-running test of 560 women between ages 30 and 70 revealed a steady decrease in maximal running velocity of 8.5% per decade (0.85% per year).²⁹ The patterns of change are about the same in sprint- and endurance-running performances.

Swimming Performance

A retrospective study of freestyle performances at the U.S. Masters swimming championships between 1991 and 1995 revealed that both men's and women's performances in the 1,500 m declined steadily from age 35 to about 70 years, after which swimming times slowed at a faster rate.³⁴ However, the rate and magnitude of the declines in both 50 m and 1,500 m performances with age were found to be greater for women than for men.

Cycling Performance

As with other strength and endurance sports, record-setting cycling performances are generally achieved in the age range of 25 to 35. Male and female cyclists' records for 40 km (24.9 mi) races decrease at about the same rate with age, an average of 20 s (approximately 0.6%) per year. The U.S. national cycling records for 20 km (12.4 mi) show a similar pattern for both men and women. For this distance, speed decreases by about 12 s (approximately 0.7%) per year from age 20 to nearly age 65.

In Review

- As we age, peak performances in both endurance and strength events decrease by about 1% to 2% per year, starting between ages 25 and 35.
- Records in running, swimming, cycling, and weightlifting indicate that we are in our physical and physiological prime during our 20s and early 30s.
- In all of these sports, performance generally declines with aging beyond age 30 to 35.
- Most athletic performances decline steadily during middle and older age, primarily because of decrements in endurance and strength.

Weightlifting

In general, maximal muscle strength is achieved between the ages of 25 and 35. Beyond that age range, as shown in [figure 18.13](#), international records for the sum of two power lifts decline at a steady rate. Of course, as with other measurements of human performance, individual strength performances vary considerably. Some individuals, for example, exhibit greater strength at age 60 than people half their age.

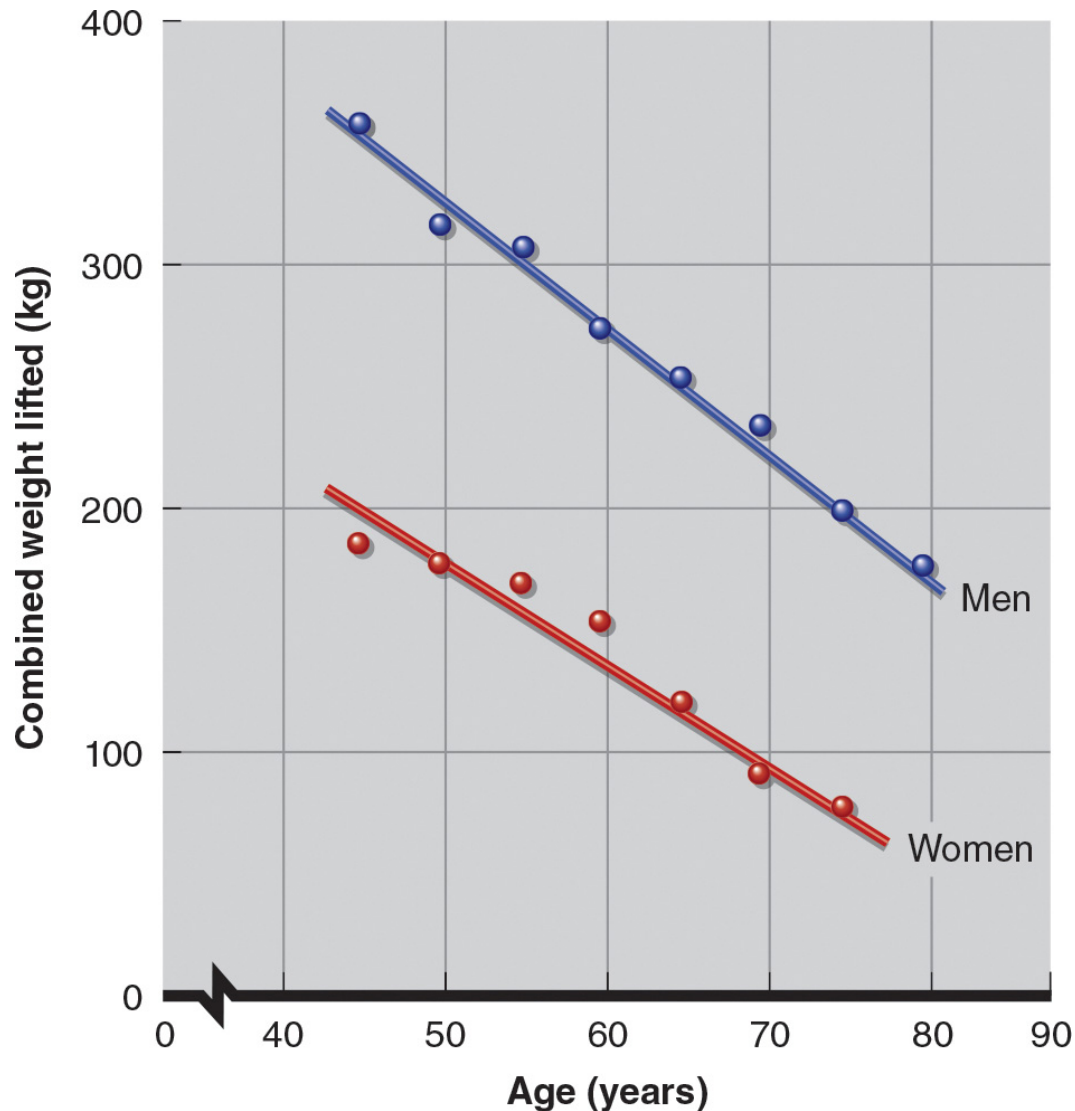


FIGURE 18.13 Changes in International Weightlifting Federation records with age among male and female weightlifters. The values reported are combined weight totals for the snatch and the clean and jerk.

Most athletic performances decline at a steady rate during middle and advanced ages. These decreases result from decrements in

both muscular and cardiovascular endurance and strength, as discussed earlier in this chapter.

Special Issues

As we age, we need to consider several special issues that can directly affect us when exercising or performing various sport activities. Here we look briefly at environmental stresses and then consider the issues of longevity, injury, and risk of death resulting from exercise and sport.

Environmental Stress

Because a variety of physiological control processes become less effective with aging, we can logically assume that older people are less tolerant of environmental stress than their younger counterparts. As we have seen elsewhere in this chapter, it is difficult to determine the separate effects of aging and physical fitness. In the following discussion, we compare the responses of younger and older adults to exercise in the heat and comment on cold stress and altitude exposure in older athletes.

Exposure to Heat

Exposure to heat stress presents a problem for older people. A preponderance of deaths during environmental heat waves occurs in people over the age of 70. The rate of metabolic heat production is related to the absolute exercise intensity, while heat loss mechanisms are related to the relative exercise intensity, so matching young and older subjects for $\dot{V}O_{2\max}$ is important. When subjects are matched for body composition and $\dot{V}O_{2\max}$, there is no difference in core temperature during exercise in the heat. However, when older subjects with a normal $\dot{V}O_{2\max}$ for their age are compared to young subjects, they have a higher core temperature (see [figure 18.14](#)).²³

VIDEO 18.4 Presents Lacy Alexander on the effects of age on thirst and temperature regulation.



These results indicate that physical training affects certain thermoregulatory responses. Sweat gland density does not appear to decline with aging but sweat gland output does, and research indicates that sweating rate is much more closely related to $\dot{V}O_{2\max}$ than it is to age. As we saw in [chapter 12](#), an increase in skin blood flow is necessary for the transfer of heat from the core to the shell for dissipation via evaporation of sweat. Even when young and older subjects are matched for $\dot{V}O_{2\max}$, skin blood flow is lower in the older subjects but higher in older highly fit compared to older normally fit subjects, indicating that regular aerobic training can improve heat dissipation. Also, exercise in the heat requires significant blood flow to both the skin and exercising muscle, which is accomplished through increased cardiac output and reduced blood flow to renal and splanchnic regions. This redistribution of blood flow is less effective in older individuals, but similar to the situation with skin blood flow, improving aerobic fitness can improve this response. One study demonstrated that 4 weeks of endurance training in older men, which improved $\dot{V}O_{2\max}$ by ~25%, improved regional redistribution so that renal and splanchnic blood flow decreased ~200 ml more than before training, which reduced cardiovascular strain.

In short, although aging reduces our ability to adapt to exercise in the heat, this deficit is largely due to reduced aerobic capacity. Increasing aerobic capacity can increase skin blood flow, increase sweating rate by increasing sweat gland output, and improve redistribution of blood to the skin and exercising muscles.

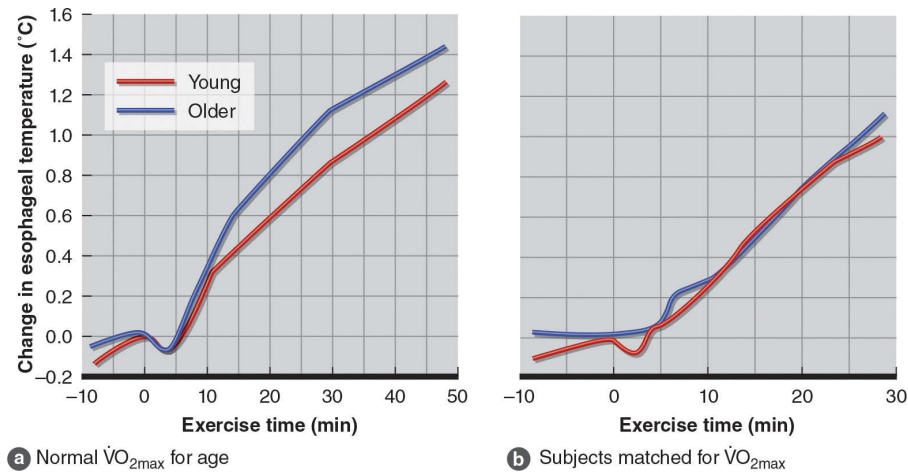


FIGURE 18.14 Changes in body core temperature in response to young (red line) and older (blue line) subjects exercising in a warm environment. When subjects in each age group have normal fitness levels for their age (a), core temperature rises more sharply in the older individuals. However, when the two age groups are selected to have the same $\dot{V}O_{2max}$, the difference in core temperature disappears (b), suggesting that $\dot{V}O_{2max}$ is more important than chronological age in determining this response.

Adapted by permission from W.L. Kenney, "Thermoregulation at Rest and During Exercise in Healthy Older Adults," *Exercise and Sport Sciences Reviews* 25 (1997): 41-77.

Exposure to Cold and to Altitude

In contrast to heat exposure, exercise in the cold typically poses less of a health risk. Due to declining aerobic fitness and loss of muscle mass, older individuals do have a reduced ability to generate metabolic heat. Also, the ability of the cutaneous vasculature to constrict is impaired with aging, which may increase heat loss. As a result of these changes, older individuals fail to appropriately maintain core temperature during cold stress. This is the case even when elderly subjects are exposed to what we would consider a mild cold stress.⁷ However, people can easily offset these decrements by wearing clothing that is appropriate for the environmental conditions and activity level. By performing so-called behavioral thermoregulation, older athletes can offset the decrements in physiological thermoregulation and continue to exercise safely in cold environments.

During exposure to high altitude, there is little reason to expect older athletes to respond differently than their younger counterparts. Unfortunately, data are lacking regarding aging and the rate and magnitude of acclimatization to altitude. Likewise, it is unclear

whether aging per se increases the incidence of any of the altitude illnesses. We can expect the performance of an older athlete at altitude to be similar to that of a younger athlete of comparable physical fitness.

Longevity

Regular physical activity is an important contributor to good health. Does training throughout adulthood affect longevity? Because the aging rate in rats is more rapid than in humans, they have been used as subjects in studies conducted to determine the influence of chronic exercise (training) on **longevity** (the length of one's life). A study by Goodrick¹¹ demonstrated that rats that exercised freely lived about 15% longer than sedentary rats. But an investigation at Washington University in St. Louis showed no significant increase in the life span of rats that voluntarily ran on an exercise wheel.¹⁷ More of the active rats lived to old age, but on average, they still died at the same age as their sedentary counterparts. It is interesting that rats that had restricted food intake and maintained a lower body weight lived 10% longer than the freely eating, sedentary rats. Although exercise training is a key component to energy balance, the only known way to increase longevity is through caloric restriction.

Of course, we cannot directly apply these findings to humans, but these results raise some interesting questions that might be relevant to our health and longevity. Although it is true that an endurance exercise program can reduce a number of the risk factors associated with cardiovascular disease, only limited information supports the contention that people will live longer if they exercise regularly. Data collected from the alumni at Harvard University and the University of Pennsylvania and from participants at the Aerobic Center in Dallas suggest that there is a decrease in mortality rate and a small increase in longevity (about 2 years) among people who remain physically active throughout life. At a minimum, regular physical activity can increase the **health span**—the number of years of generally healthy living, free from serious disease or chronic disability.

- The impaired ability of older individuals to tolerate exercise in the heat is due to reduced $\dot{V}O_{2\max}$ and impaired cardiovascular adaptations rather than a direct effect of aging on thermoregulatory control or sweating.
 - Regular exercise training can increase skin blood flow and sweating rate and improve the redistribution of cardiac output in older individuals as well as young men and women.
 - Older people generally have an impaired ability to tolerate cold, but they can compensate by wearing appropriate clothing.
 - Adaptation to altitude appears to be independent of age.
 - An active lifestyle appears to be associated with a small increase in longevity. Just as important, an active lifestyle leads to a higher quality of life!
 - There is an increased risk of injury from exercise as people age, and injuries tend to be slower to heal.
 - The risk of death during exercise is not increased in those who are regularly active but is increased in those who seldom exercise.
-

IN CLOSING

In this chapter, we examined the effects of aging on physical performance. We evaluated changes in cardiorespiratory endurance and strength with age. We considered the effect of aging on body composition, which we know can affect performance. And yet, in the course of our discussion, it became clear that much of the change that occurs with aging is attributable to the inactivity that often accompanies aging. When older people participate in training, most of the changes associated with aging are lessened, and the resulting degree of change is similar to that seen in young and middle-aged adults. Thus, we have dispelled many of the myths about the capacity for physical activity of older people.

In the next chapter, we turn our attention to women, who as a group are often considered less capable of physical activity than men. We consider the physiology of girls and women, the impact of this physiology on athletic ability, how performances of female athletes compare with those of male athletes, and special issues associated with being female.

KEY TERMS

cardiovascular deconditioning

endothelial dysfunction

forced expiratory volume in 1 s ($FEV_{1.0}$)

health span

longevity

maximal expiratory ventilation ($\dot{V}O_{2max}$)

osteopenia

osteoporosis

peripheral blood flow

primary aging

STUDY QUESTIONS

1. What changes occur in height, weight, and body composition with aging? What accounts for these changes? How do these changes affect maximal oxygen uptake?
2. What changes occur in muscle with aging? How do they affect strength and athletic performance?
3. Describe the changes in HR_{max} with age. How does training alter this relationship?

4. How does aging affect maximal stroke volume and maximal cardiac output? What mechanisms can potentially explain these changes?
5. How does the respiratory system change with aging? What happens to vital capacity, FEV_{1.0}, residual volume, total lung capacity, and the ratio RV/TLC?
6. $\dot{V}O_{2max}$ declines with age across the entire population. Describe the physiological mechanisms that account for this decline. How do trained older individuals maintain a relatively high $\dot{V}O_{2max}$?
7. How do aging and habitual exercise affect blood vessel function?
8. How does age affect anaerobic function?
9. Differentiate between biological aging and physical inactivity.
10. What influence do aging and training have on body composition?
11. Describe the trainability of the older individual for both strength and aerobic endurance.
12. What can be done to minimize age-related losses in motor function and mobility?
13. Describe the changes in strength and endurance performance records with aging.
14. What concerns should we have about older people exercising in hot and cold environments or at altitude?
15. What is the difference between life span and health span?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.

